

PalmSource®

Conference and Expo



Optimum Hacking

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Optimum hacking.
Settle for nothing less.



What do you mean by "Hack"?

- Things that run under HackMaster
- Fragile, ugly code that is lucky to work at all
- Code that does really cool unexpected tricks, going beyond expectations
- Things that use undocumented APIs, mess with private data structures, do things they're not supposed to do



The dilemmas

- Hacks are good
- Hacks are bad

- Hacks make sales
- Hacks break too easily

- Hacks fill needs
- Hacks are hard to use



Hacks aren’t a matter of morals

- There's nothing wrong with hacking!
- There's just a consequence down the road when things change and the hack doesn't work
- It isn't a moral decision; it is a business decision
  - Balance the quick fix against the future effort of having to fix it a second (or third) time.
  - Or, balance the “coolness” factor and immediate sales, against future unhappy users
  - Also consider the danger to the user’s data
  - Also consider support costs



HackMaster (and its successors)

- Unified UI, basic rules and organization
- Lets users manage hacks.
- Lets users uninstall hacks without rebooting
- Very helpful software! Don't patch without it!
- Won't work on Palm OS® Version 5 or beyond



Assumptions

- Every hack needs to make some assumptions
- Good hacks check their assumptions
  - sysFtrNumProcessorID
  - sysFtrNumROMVersion
  - Error return codes
  - NULL pointers
- Differentiates between "Carefully targeted" hacks and "Hopelessly fragile" ones



More Checking

- Other good candidates to check:
  - sysFtrNumOEMCompanyID
  - sysFtrNumOEMDeviceID
  - sysFtrNumOEMHALID



Hacking on Screens

- High-density
- Different size screens
- Other complicating factors
  - Graphics accelerators
  - VRAM
- Offscreen buffers
  - Usually safe to mess with
- Bitmaps on PACE (shadow structures)



Popular Hacking targets

- The screen
- The processor
- Special hardware
- Tasks
- UI objects
- Various preferences and data
- Various traps



Hacking on Palm OS 5

- Lots of new opportunities!
  - (Change is threatening!)
- Pace and shadow structures
- Notifications
- Simulator versus ARM
- DAL (and HAL)



Hacking In The Future

- Next major OS version affects hacking
  - Protected memory spaces
  - Security enhancements
  - New patching mechanism
- Good hacks should keep working
- Devices are becoming more diverse



Suggestions for Hacking

- Overlays and Database tricks
- VFS
- Head-patching event handlers
- Notifications
  - Lots of new ones in Palm OS Version 5
  - (Someone should do a HackMaster hack for these)



Best Tools

- Lots of devices, lots of ROMs
- Pose
- PalmDebugger
- Palm Reporter
- Palm OS® Version 5 Simulator
  - Because of new debugging features



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